PROBLEM 1: Student Grade Calculator

```
import java.io.*;

class Marks{

public static void main(String args[]) throws IOException

{

BufferedReader br = new BufferedReader (new InputStreamReader (System.in));

System.out.println("Enter marks in subject 1:");

int s1 = Integer.parseInt(br.readLine());


System.out.println("Enter marks in subject 2:");

int s2 = Integer.parseInt(br.readLine());


System.out.println("Enter marks in subject 3:");

int s3 = Integer.parseInt(br.readLine());


System.out.println("Enter marks in subject 4:");

int s4 = Integer.parseInt(br.readLine());


System.out.println("Enter marks in subject 5:");

int s5 = Integer.parseInt(br.readLine());


int total;

total = s1+s2+s3+s4+s5;

System.out.println("\ntotal marks="+ total);
```

```
float percentage;

percentage = (float) total/500*100;

System.out.println("Percentage ="+ percentage +"%");


if (percentage>=90){

System.out.println("Grade A+");

System.out.println("Student has Passed");


}

else if (percentage>=80 && percentage<=89){

System.out.println("Grade A");

System.out.println("Student has Passed");

}

else if (percentage>=70 && percentage<=79){

System.out.println("Grade B");

System.out.println("Student has Passed");

}

else if (percentage>=60 && percentage<=69){

System.out.println("Grade C");

System.out.println("Student has Passed");

}

else if (percentage>=50 && percentage<=59){

System.out.println("Grade D");
```

```
System.out.println("Student has Passed");
}
else if (percentage<50){
System.out.println("Fail");
}
}
}
```

PROBLEM 2: Electricity Bill Generator

```
import java.io.*;

class Bill{

public static void main(String args[]) throws IOException

{


BufferedReader br = new BufferedReader (new InputStreamReader (System.in));

System.out.println("Enter consumer's name:");

String name = (br.readLine());


System.out.println("Enter consumer's number:");

int number = Integer.parseInt(br.readLine());


System.out.println("Enter units consumed:");

int units = Integer.parseInt(br.readLine());


double bill = 0;

double surcharge = 0;

double totalBill;


 if (units <= 100)

        {

            bill = units * 3;

        }
```

```java
else if (units <= 200)
{
    bill = (100 * 3) + ((units - 100) * 5);
}
else if (units <= 500)
{
    bill = (100 * 3) + (100 * 5) + ((units - 200) * 7);
}
else
{
    bill = (100 * 3) + (100 * 5) + (300 * 7)
           + ((units - 500) * 10);
}

// Apply 5% surcharge if bill exceeds ₹5000
if (bill > 5000)
{
    surcharge = bill * 0.05;
}

totalBill = bill + surcharge;

// Display details
System.out.println("\nElectricity Bill");
```

```java
        System.out.println("Consumer Name   : " + name);
        System.out.println("Consumer Number : " + number);
        System.out.println("Units Consumed  : " + units);
        System.out.println("Bill Amount     : rs." + bill);
        System.out.println("Surcharge       : rs." + surcharge);
        System.out.println("Total Bill      : rs." + totalBill);
    }
}
```

PROBLEM 4: Array Statistics

```
import java.io.*;

class Statistics

{

    public static void main(String args[]) throws IOException

    {

        BufferedReader br = new BufferedReader(new InputStreamReader(System.in));



        int[] arr = new int[10];



        int max, min;

        int sum = 0;

        int even = 0;

        int odd = 0;



                System.out.println("Enter 10 integers:");



        for(int i = 0; i < 10; i++)

        {

            arr[i] = Integer.parseInt(br.readLine());

        }




        max = arr[0];
```

```
min = arr[0];



for(int i = 0; i < 10; i++)
{
    sum = sum + arr[i];

    if(arr[i] > max)
        max = arr[i];

    if(arr[i] < min)
        min = arr[i];

    if(arr[i] % 2 == 0)
        even++;
    else
        odd++;
}

double average = sum / 10.0;



System.out.println("Maximum = " + max);
System.out.println("Minimum = " + min);
```

```
System.out.println("Average = " + average);
System.out.println("Even numbers = " + even);
System.out.println("Odd numbers = " + odd);



System.out.print("Enter number to search: ");
int key = Integer.parseInt(br.readLine());

boolean found = false;

for(int i = 0; i < 10; i++)
{
    if(arr[i] == key)
    {
        found = true;
        break;
    }
}

if(found)
    System.out.println("Number found.");
else
    System.out.println("Number not found.");
```

```
        System.out.println("Array in reverse order:");

        for(int i = 9; i >= 0; i--)
        {
            System.out.print(arr[i] + " ");
        }
    }
}
```

PROBLEM 3: Number Analysis Tool

```
import java.io.*;

public class Problem3{

    public static void main (String args[]) throws IOException {

      BufferedReader br = new BufferedReader(new InputStreamReader(System.in)) ;

      System.out.println(" enter the value of N: ") ;

       int n = Integer.parseInt(br.readLine()) ;

                System.out.println("1.Check whether the number is Prime  ") ;

                 System.out.println("2.Check whether it is Palindrome ");

                  System.out.println("3.Check whether it is Armstrong ");

                    System.out.println("4.Find the Sum of Digits");

                      System.out.println("5.Reverse the Number");

                       System.out.println("6.Exit");

                  System.out.print("Enter your choice: ");

        int x = Integer.parseInt(br.readLine());

       int num = n;

         int  count = 0 ;

         switch(x) {

             case 1:

                for ( int i = 1; i <=n ; i++){

                if ( n % i == 0 ) count++;

              }

                 if ( count == 2 ) System.out.println("The number is prime") ;
```

```
    else System.out.println(" The number is not prime");
  break;
  case 2:
    if ( n < 0 ) System.out.println(" Negative numbers are not allowed ") ;
  int  rev = 0;
   int No = n;
    while(No != 0) {
 rev = rev * 10 + No % 10;
 No /= 10;
}
System.out.println("The number is palindrome");
break ;
case 3:
  int sum3 =0 ;
  int digits = String.valueOf(num).length();
  while (num > 0) {
 int lastDigit = num % 10;
 sum3 += Math.pow(lastDigit, digits);
 num /= 10;
}
if(sum3 == num) System.out.println("The number is an Armstrong " ) ;
 else System.out.println("The number is not an Armstrong");
 break;
 case 4:
```

```
int temp4 ;
int sum4 =0 ;
while ( num > 0){
  temp4 = num %10;
 sum4 = sum4 + temp4 ;
 num /= 10;
}
 System.out.print(sum4);
break;
case 5:
int Rev = 0 ;
int temp5;
 num = n ;
while (num > 0){
  temp5 = num %10 ;
   Rev = ( Rev*10) + temp5;
   num /= 10;
}
 System.out.println("The reverse of the number is : "+Rev);
 break;
 case 6:
   break;
   default:
     System.out.println("Invalid choice");
```

```
        }
      }
    }
```

PROBLEM 5: **Employee Salary Management System**

```
import java.io.*;

import java.util.Arrays;

public class Problem5 {

   public static void main (String[] args) throws IOException {

      BufferedReader br = new BufferedReader(new InputStreamReader(System.in));

      System.out.print(" Enter the number of employees : ");

      int n = Integer.parseInt( br.readLine() );

      int[] arr = new int[n] ;

      int i ;

      System.out.println("Enter the ID of  employees : ");

      for(i=0;i<n;i++) {

         arr[i]=Integer.parseInt(br.readLine());

      }

      String[] name = new String[n];

      System.out.println("Enter the names of employees : ") ;

      for(i=0;i<n;i++){

      name[i] = br.readLine();

      }

        int[] DA = new int[n];

         int[] HRA =new int[n];

         int[] PF =new int[n];

         int[] gross = new int[n];

         int[] net_salary = new int[n];
```

```
int sum  = 0;
    System.out.println("Enter the basic salary of the employees : " ) ;
int[] basic_salary = new int[n];
 for( i = 0 ;i< n; i++) {
   basic_salary[i] = Integer.parseInt(br.readLine());
   DA[i] = (40 * basic_salary[i]) / 100;
   HRA[i] = (20 * basic_salary[i]) / 100;
   PF[i] = (12 * basic_salary[i]) / 100;
   gross[i] = basic_salary[i] + DA[i] + HRA[i];
   net_salary[i] = gross[i] - PF[i];
   sum += net_salary[i];
   }
         System.out.println(" The names are : "+Arrays.toString(name));
   for ( i=0;i<n;i++){
      System.out.println("The ID of the employee is : "+arr[i]);


   System.out.println("Salary for " + name[i] + " is: " + basic_salary[i]);
   }
   int highestIndex =0;
   int lowestIndex =0;
   for ( i=1;i<n;i++){
     if ( basic_salary[i]>highestIndex)highestIndex= i ;
      else if( basic_salary[i] < lowestIndex ) lowestIndex = i;
      }
```

```
        System.out.println("\n Employee Details");
    for (i = 0; i < n; i++){
      System.out.println("ID: " + arr[i] + "  Name: " + name[i] +  "  Basic Salary: " + basic_salary[i] + "  Net Salary : " + net_salary[i]);
    }
    int average = sum / n;
    System.out.println("\n Summary ");
    System.out.println("Highest Paid : " + name[highestIndex] + " (" + net_salary[highestIndex] + ")");
    System.out.println("Lowest Paid  : " + name[lowestIndex] + " (" + net_salary[lowestIndex] + ")");
    System.out.println("Average Salary: " + average);
      }
        }
```